

Path 2: Cyber Security 101 > Module 5: Networking > Room 1: Networking Concepts

<https://tryhackme.com/room/networkingconcepts>

Networking Concepts:

Learn about the ISO OSI model and the TCP/IP protocol suite.

>> Task 1: Introduction

Have you ever wondered why you need an IP address to access the Internet? Is it true that an IP address can uniquely identify the user? Are you curious to learn what the life of a packet looks like? If the answer is yes, let's dive in!

♦ Room Prerequisites:

This room expects that you know terms such as IP address and TCP port number; however, we don't expect that the reader is able to explain such terms in proper technical depth. If you are unfamiliar with these terms, please consider joining the "Pre Security" path.

♦ Learning Objectives:

By the time you finish this room, you will have learned about the following:

- ISO OSI network model
- IP addresses, subnets, and routing
- TCP, UDP, and port numbers
- How to connect to an open TCP port from the command line

>> Task 2: OSI Model

The OSI (Open Systems Interconnection) model is a conceptual model developed by the International Organization for Standardization (ISO) that describes how communications should occur in a computer network. In other words, the OSI model defines a framework for computer network communications. Although this model is theoretical, it is vital to learn and understand as it helps grasp networking concepts on a deeper level. The OSI model is composed of seven layers:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Hint to remember: **"Please Do Not Throw Spinach Pizza Away". "Please Do Nice Things, Smile Politely Always".**

The numbering starts with the physical layer being layer 1, while the top layer, the application layer, is layer 7. To help you remember the layers from bottom to top, you can use a mnemonic such as "Please Do Not Throw Spinach Pizza Away." You can check the Internet for other easy-to-remember acronyms if this helps you memorise them. Remembering the OSI model layers with their layer numbers is important; otherwise, you will struggle to understand terms such as "layer 3 switch" or "layer 7 firewall."

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♦ Layer 1: Physical Layer

The physical layer, also referred to as layer 1, deals with the physical connection between devices; this includes the medium, such as a wire, and the definition of the binary digits 0 and 1. Data transmission can be via an electrical, optical, or wireless signal. Consequently, we need data cables or antennas, depending on our physical medium.

In addition to Ethernet cable and optical fibre cable, examples of the physical layer medium include the WiFi radio bands, the 2.4 GHz band, the 5 GHz band, and the 6 GHz band.

Physical Layer (Layer 1)

What is the Physical Layer?

- The Physical Layer is the lowest layer of the OSI model. It deals with sending raw data as signals (0s and 1s).

What does this layer do?

- Sends and receives electrical, optical, or wireless signals
- Defines cables, connectors, voltage, and speed

Purpose of it:

- ✓ Move data physically
- ✓ Provide a path for communication

Day-to-day life example:



Road used by vehicles

Without a road, vehicles can't move.

Real technical example:

Ethernet cable

Fiber optic cable

Wi-Fi signals

One-line summary:

Physical Layer transfers raw data bits through cables or wireless signals.

♦ Layer 2: Data Link Layer

The physical layer defines a medium to transmit our signal. The data link layer, i.e., layer 2, represents the protocol that enables data transfer between nodes on the same network segment. Let's put it in simpler terms. The data link layer describes an agreement between the different systems on the same network segment on how to communicate. A network segment refers to a group of networked devices using a shared medium or channel for information transfer. For example, consider a company office with ten computers connected to a network switch; that's a network segment.

Examples of layer 2 include Ethernet, i.e., 802.3, and WiFi, i.e., 802.11. Ethernet and WiFi addresses are six bytes. Their address is called a MAC address, where MAC stands for Media Access Control. They are usually expressed in hexadecimal format with a colon separating each two hexadecimal digits (one byte). The three leftmost bytes identify the vendor.

a4:c3:f0:85:ac:2d (MAC Address)

Vendor who build the network interfere, in this instance, "Intel": **a4:c3:f0**

Unique address of the network interface: **85:ac:2d**

We expect to see two MAC addresses in each frame in real network communication over Ethernet or WiFi. The packet in the screenshot below shows:

- The destination data-link address (MAC address) highlighted in yellow
- The source data link address (MAC address) is highlighted in blue
- The remaining bits show the data being sent

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What is the Data Link Layer?

- The Data Link Layer ensures error-free delivery of data between directly connected devices.

What does this layer do?

- Uses MAC addresses
- Detects and fixes transmission errors
- Controls data flow

Purpose of it:

- ✓ Reliable local communication
- ✓ Identify devices in the same network

Day-to-day life example:



Courier delivering parcel inside a building using flat number

Real technical example:

Switch

Network Interface Card (NIC)

Wi-Fi access point

One-line summary:

Data Link Layer ensures correct data delivery within the same network using MAC addresses.

♦ Layer 3: Network Layer

The data link layer focuses on sending data between two nodes on the same network segment. The network layer, i.e., layer 3, is concerned with sending data between different networks. In more technical terms, the network layer handles logical addressing and routing, i.e., finding a path to transfer the network packets between the diverse networks.

In the data link layer, we gave an example of one company office with ten computers, where the data link layer is responsible for providing a connection between them. Let's say that this company has multiple offices distributed across various cities, countries, or even continents. The network layer is responsible for connecting the different offices together.

The network below shows that computers A and B are connected, although on different networks. You can also notice two paths connecting the two computers; the network layer will route the network packets through the path it deems better.

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What is the Network Layer?

- The Network Layer decides where data should go and which path to take.

What does this layer do?

- Uses IP addresses
- Performs routing
- Selects best path

Purpose of it:

- ✓ Connect different networks
- ✓ Ensure data reaches the correct destination

Day-to-day life example:

 Google Maps choosing the best route from one city to another

Real technical example:

Routers
IP routing

One-line summary:

Network Layer routes data between different networks using IP addresses.

♦ Layer 4: Transport Layer

Layer 4, the transport layer, enables end-to-end communication between running applications on different hosts. Your web browser is connected to the TryHackMe web server over the transport layer, which can support various functions like flow control, segmentation, and error correction.

Examples of layer 4 are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).

What is the Transport Layer?

- The Transport Layer ensures data is delivered correctly and completely.

What does this layer do?

- Breaks data into segments
- Uses TCP or UDP
- Handles retransmission

Purpose of it:

- ✓ Reliable data delivery
- ✓ Maintain data order

Day-to-day life example:

📦 Sending multiple boxes and checking all boxes arrive

Real technical example:

TCP (reliable – email, web)

UDP (fast – video, gaming)

One-line summary:

Transport Layer ensures data reaches the destination correctly and in order.

♦ Layer 5: Session Layer

The session layer is responsible for establishing, maintaining, and synchronising communication between applications running on different hosts. Establishing a session means initiating communication between applications and negotiating the necessary parameters for the session. Data synchronisation ensures that data is transmitted in the correct order and provides mechanisms for recovery in case of transmission failures.

Examples of the session layer are Network File System (NFS) and Remote Procedure Call (RPC).

What is the Session Layer?

- The Session Layer manages start, maintain, and end of communication sessions.

What does this layer do?

- Opens sessions
- Keeps sessions alive
- Closes sessions properly

Purpose of it:

- ✓ Maintain continuous communication
- ✓ Prevent session interruption

Day-to-day life example:

 Phone call start → talk → end

Real technical example:

Login session

Video call session

One-line summary:

Session Layer manages communication sessions between devices.

♦ Layer 6: Presentation Layer

The presentation layer ensures the data is delivered in a form the application layer can understand. Layer 6 handles data encoding, compression, and encryption. An example of encoding is character encoding, such as ASCII or Unicode.

Various standards are used at the presentation layer. Consider the scenario where we want to send an image via email. First, we use JPEG, GIF, and PNG to save our images; furthermore, although hidden from the user by the email client, we use MIME (Multipurpose Internet Mail Extensions) to attach the file to our email. MIME encodes a binary file using 7-bit ASCII characters.

What is the Presentation Layer?

- The Presentation Layer prepares data so the receiver can understand and trust it.

What does this layer do?

- Encrypts and decrypts data
- Formats and compresses data

Purpose of it:

- ✓ Data security
- ✓ Data readability

Day-to-day life example:

 Locking a letter in a secret code

Real technical example:

HTTPS encryption

File formats (JPEG, MP3)

One-line summary:

Presentation Layer encrypts and formats data for secure communication.

♦ Layer 7: Application Layer

The application layer provides network services directly to end-user applications. Your web browser would use the HTTP protocol to request a file, submit a form, or upload a file.

The application layer is the top layer, and you might have encountered many of its protocols as you use different applications. Examples of Layer 7 protocols are HTTP, FTP, DNS, POP3, SMTP, and IMAP. Don't worry if you are not familiar with all of them.

What is the Application Layer?

- The Application Layer is where users interact with the network.

What does this layer do?

- Provides network services to applications
- Enables user communication

Purpose of it:

- ✓ Allow users to access network services

Day-to-day life example:

 Writing and sending a letter

Real technical example:

Web browser
Email
FTP

One-line summary:

Application Layer allows users to communicate over the network.

♦ Summery:

Reading about the ISO OSI model for the first time can be intimidating; however, it becomes easier as you progress in your study of networking protocols. To help with your studies, we have summarised the ISO OSI layers in the table below.

Layer Number	Layer Name	Main Function	Example Protocols and Standards
Layer 7	Application Layer	Providing services and interfaces to applications	HTTP, FTP, DNS, POP3, SMTP, IMAP
Layer 6	Presentation Layer	Data encoding, encryption, and compression	Unicode, MIME, JPEG, PNG, MPEG
Layer 5	Session Layer	Establishing, maintaining, and synchronising sessions	NFS, RPC
Layer 4	Transport Layer	End-to-end communication and data segmentation	UDP, TCP
Layer 3	Network Layer	Logical addressing and routing between networks	IP, ICMP, IPSec
Layer 2	Data Link Layer	Reliable data transfer between adjacent nodes	Ethernet (802.3), WiFi (802.11)
Layer 1	Physical Layer	Physical data transmission media	Electrical, optical, and wireless signals

Q1) Which layer is responsible for end-to-end communication between running applications?

Answer: 4

Q2) Which layer is responsible for routing packets to the proper network?

Answer: 3

Q3) In the OSI model, which layer is responsible for encoding the application data?

Answer: 6

Q4) Which layer is responsible for transferring data between hosts on the same network segment?

Answer: 2

>> Task 3: TCP/IP Model

Now that we have covered the conceptual ISO OSI model, it is time to study an implemented model, the TCP/IP model. TCP/IP stands for Transmission Control Protocol/Internet Protocol and was developed in the 1970s by the Department of Defense (DoD). I hear you ask why DoD would create such a model. One of the strengths of this model is that it allows a network to continue to function as parts of it are out of service, for instance, due to a military attack. This capability is possible in part due to the design of the routing protocols to adapt as the network topology changes.

In our presentation of the ISO OSI model, we went from bottom to top, from layer 1 to layer 7. In this task, let's look at things from a different perspective, from top to bottom. From top to bottom, we have:

- **Application Layer:** The OSI model application, presentation and session layers, i.e., layers 5, 6, and 7, are grouped into the application layer in the TCP/IP model.
- **Transport Layer:** This is layer 4.
- **Internet Layer:** This is layer 3. The OSI model's network layer is called the Internet layer in the TCP/IP model.
- **Link Layer:** This is layer 2.

The table below shows how the TCP/IP model layers map to the ISO/OSI model layers.

Layer Number	ISO OSI Model	TCP/IP Model (RFC 1122)	Protocols
7	Application Layer	Application Layer	HTTP, HTTPS, FTP, POP3, SMTP, IMAP, Telnet, SSH
6	Presentation Layer		
5	Session Layer		
4	Transport Layer	Transport Layer	TCP, UDP
3	Network Layer	Internet Layer	IP, ICMP, IPSec

2	Data Link Layer	Link Layer	Ethernet 802.3, WiFi 802.11
1	Physical Layer		

Many modern networking textbooks show the TCP/IP model as five layers instead of four. For example, in *Computer Networking: A Top-Down Approach* 8th Edition, Kurose and Ross (https://gaia.cs.umass.edu/kurose_ross/index.php) describe the following five-layer Internet protocol stack by including the physical layer:

- Application
- Transport
- Network
- Link
- Physical

Q1) To which layer does HTTP belong in the TCP/IP model?

Answer: Application Layer

Q2) How many layers of the OSI model does the application layer in the TCP/IP model cover?

Answer: 3

>> Task 4: IP Addresses and Subnets

When you hear the word IP address, you might think of an address like **192.168.0.1** or something less common, such as **172.16.159.243**. In both cases, you are right. Both of these are IP addresses; IPv4 (IP version 4) addresses to be specific.

Every host on the network needs a unique identifier for other hosts to communicate with him. Without a unique identifier, the host cannot be found without ambiguity. When using the TCP/IP protocol suite, we need to assign an IP address for each device connected to the network.

One analogy of an IP address is your home postal address. Your postal address allows you to receive letters and parcels from all over the world. Furthermore, it can identify your home without ambiguity; otherwise, you cannot shop online!

As you might already know, we have IPv4 and IPv6 (IP version 6). IPv4 is still the most common, and whenever you come across a text mentioning IP without the version, we expect them to mean IPv4.

So, what makes an IP address? An IP address comprises four octets, i.e., 32 bits. Being 8 bits, an octet allows us to represent a decimal number between 0 and 255. An IP address is shown in the image below.

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192.168.1.1 (4 Octet with 0-255 Range) (1 Octet = 1 Byte = 8 Bits)

At the risk of oversimplifying things, the 0 and 255 are reserved for the network and broadcast addresses, respectively. In other words, **192.168.1.0** is the network address, while **192.168.1.255** is the broadcast address. Sending to the broadcast address targets all the hosts on the network. With simple math, you can conclude that we cannot have more than 4 billion unique IPv4 addresses. If you are curious about the math, it is approximately 232 because we have 32 bits. This number is approximate because we didn't consider network and broadcast addresses.

♦ Looking Up Your Network Configuration:

You can look up your IP address on the MS Windows command line using the command **ipconfig**. On Linux and UNIX-based systems, you can issue the command **ifconfig** or **ip address show**, which can be typed as **ip a s**. In the terminal window below, we show **ifconfig**.

Example Command: `ifconfig`

The terminal output above indicates the following:

- The host (laptop) IP address is **192.168.66.89**
- The subnet mask is **255.255.255.0**
- The broadcast address is **192.168.66.255**

Let's use `ip a s` to compare how the network card IP address is presented.

Command: `ip a s`

The terminal output above indicates the following:

- The host (laptop) IP address is **192.168.66.89/24**
- The broadcast address is **192.168.66.255**

If you are wondering, a subnet mask of **255.255.255.0** can also be written as **/24**. The **/24** means that the leftmost 24 bits within the IP address do not change across the network, i.e., the subnet. In other words, the leftmost three octets are the same across the whole subnet; therefore, we can expect to find addresses that range from **192.168.66.1** to **192.168.66.254**. Similar to what was mentioned earlier, **192.168.66.0** and **192.168.66.255** are the network and broadcast addresses, respectively.

♦ Private Addresses:

As we are explaining IP addresses, it is useful to mention that for most practical purposes, there are two types of IP addresses:

- **Public IP addresses**
- **Private IP addresses**

RFC 1918 defines the following three ranges of private IP addresses:

- **10.0.0.0 - 10.255.255.255 (10/8)**
- **172.16.0.0 - 172.31.255.255 (172.16/12)**
- **192.168.0.0 - 192.168.255.255 (192.168/16)**

We presented earlier an analogy stating that a public IP address is like your home postal address. A private IP address is different; the original idea is that it cannot reach or be reached from the outside world. It is like an isolated city or a compound, where all houses and apartments are numbered systematically and can easily exchange mail with each other, but not with the outside world. For a private IP address to access the Internet, the router must have a public IP address and must support Network Address Translation (NAT). At this stage, let's not worry about understanding how NAT works, as we will revisit it later in this module.

Before moving on, I recommend memorising the private IP address ranges. Otherwise, you might see an IP address such as **10.1.33.7** or **172.31.33.7** and try to access it from a public IP address.

- ♦ Routing:

A router is like your local post office; you hand them the mail parcel, and they would know how to deliver it. If we dig deeper, you might mail something to an address in another city or country. The post office will check the address and decide where to send it next. For example, if it is to leave the country, we expect one central office to handle all shipments abroad.

In technical terms, a router forwards data packets to the proper network. Usually, a data packet passes through multiple routers before it reaches its final destination. The router functions at layer 3, inspecting the IP address and forwarding the packet to the best network (router) so the packet gets closer to its destination.

Q1) Which of the following IP addresses is not a private IP address?

(192.168.250.125, 10.20.141.132, 49.69.147.197, 172.23.182.251)

Answer: 49.69.147.197

Q2) Which of the following IP addresses is not a valid IP address? (192.168.250.15, 192.168.254.17, 192.168.305.19, 192.168.199.13)

Answer: 192.168.305.19

>> Task 5: UDP and TCP

The IP protocol allows us to reach a destination host on the network; the host is identified by its IP address. We need protocols that would enable processes on networked hosts to communicate with each other. There are two transport protocols to achieve that: UDP and TCP.

♦ UDP:

UDP (User Datagram Protocol) allows us to reach a specific process on this target host. UDP is a simple connectionless protocol that operates at the transport layer, i.e., layer 4. Being connectionless means that it does not need to establish a connection. UDP does not even provide a mechanism to know that the packet has been delivered.

An IP address identifies the host; we need a mechanism to determine the sending and receiving process. This can be achieved by using port numbers. A port number uses two octets; consequently, it ranges between 1 and 65535; port 0 is reserved. (The number 65535 is calculated by the expression $2^{16} - 1$.)

A real-life example similar to UDP is the standard mail service, with no delivery confirmation. In other words, there is no guarantee that the UDP packet has been received successfully, similar to the case of sending a parcel using standard mail with no confirmation of delivery. In the case of standard mail, it means a cheaper cost than the mail delivery options with confirmation. In the case of UDP, it means better speed than a transport protocol that provides “confirmation.”

But what if we want a transport protocol that acknowledges received packets? The answer lies in using TCP instead of UDP.

♦ TCP:

TCP (Transmission Control Protocol) is a connection-oriented transport protocol. It uses various mechanisms to ensure reliable data delivery sent by the different processes on the networked hosts. Like UDP, it is a layer 4 protocol. Being connection-oriented, it requires the establishment of a TCP connection before any data can be sent.

In TCP, each data octet has a sequence number; this makes it easy for the receiver to identify lost or duplicated packets. The receiver, on the other hand, acknowledges the reception of data with an acknowledgement number specifying the last received octet.

A TCP connection is established using what's called a three-way handshake. Two flags are used: SYN (Synchronise) and ACK (Acknowledgment). The packets are sent as follows:

SYN Packet: The client initiates the connection by sending a SYN packet to the server. This packet contains the client's randomly chosen initial sequence number.

SYN-ACK Packet: The server responds to the SYN packet with a SYN-ACK packet, which adds the initial sequence number randomly chosen by the server.

ACK Packet: The three-way handshake is completed as the client sends an ACK packet to acknowledge the reception of the SYN-ACK packet.

Similar to UDP, TCP identifies the process of initiating or waiting (listening) for a connection using port numbers. As stated, a valid port number ranges between 1 and 65535 because it uses two octets and port 0 is reserved.

Q1) Which protocol requires a three-way handshake?

Answer: TOP

Q2) What is the approximate number of port numbers (in thousands)?

Answer: 65

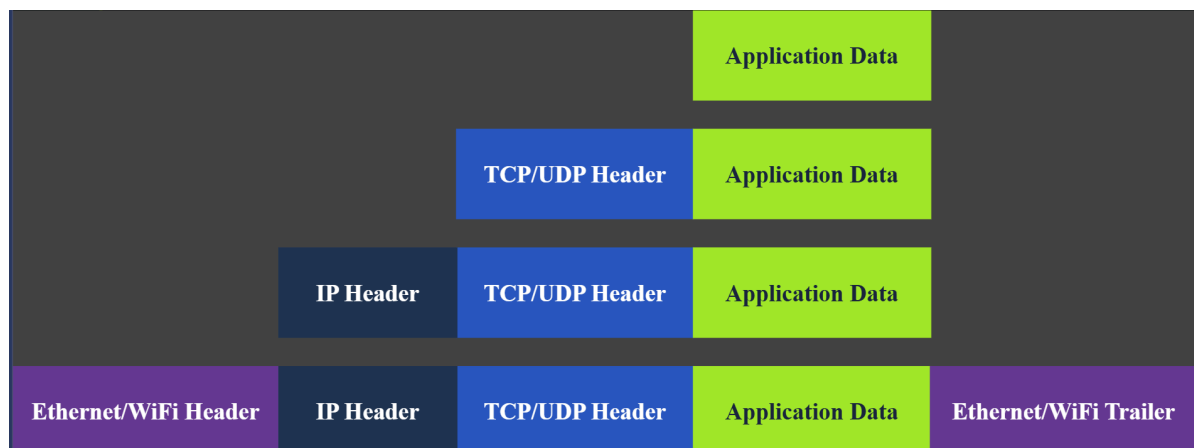
>> Task 6: Encapsulation

Encapsulation refers to the process of every layer adding a header (and sometimes a trailer) to the received unit of data and sending the **encapsulated** unit to the layer below.

Encapsulation is an essential concept as it allows each layer to focus on its intended function. In the image below, we have the following four steps:

- **Application data:** It all starts when the user inputs the data they want to send into the application. For example, you write an email or an instant message and hit the send button. The application formats this data and starts sending it according to the application protocol used, using the layer below it, the transport layer.
- **Transport protocol segment or datagram:** The transport layer, such as TCP or UDP, adds the proper header information and creates the TCP segment (or UDP datagram). This segment is sent to the layer below it, the network layer.
- **Network packet:** The network layer, i.e. the Internet layer, adds an IP header to the received TCP segment or UDP datagram. Then, this IP packet is sent to the layer below it, the data link layer.
- **Data link frame:** The Ethernet or WiFi receives the IP packet and adds the proper header and trailer, creating a frame.

We start with application data. At the transport layer, we add a TCP or UDP header to create a **TCP segment** or **UDP datagram**. Again, at the network layer, we add the proper IP header to get an **IP packet** that can be routed over the Internet. Finally, we add the appropriate header and trailer to get a WiFi or Ethernet frame at the link layer.



The process has to be reversed on the receiving end until the application data is extracted.

- ♦ The Life of a Packet:

Based on what we have studied so far, we can explain a simplified version of the packet's life. Let's consider the scenario where you search for a room on TryHackMe.

1. On the TryHackMe search page, you enter your search query and hit enter.
2. Your web browser, using HTTPS, prepares an HTTP request and pushes it to the layer below it, the transport layer.
3. The TCP layer needs to establish a connection via a three-way handshake between your browser and the TryHackMe web server. After establishing the TCP connection, it can send the HTTP request containing the search query. Each TCP segment created is sent to the layer below it, the Internet layer.
4. The IP layer adds the source IP address, i.e., your computer, and the destination IP address, i.e., the IP address of the TryHackMe web server. For this packet to reach the router, your laptop delivers it to the layer below it, the link layer.
5. Depending on the protocol, The link layer adds the proper link layer header and trailer, and the packet is sent to the router.
6. The router removes the link layer header and trailer, inspects the IP destination, among other fields, and routes the packet to the proper link. Each router repeats this process until it reaches the router of the target server.

The steps will then be reversed as the packet reaches the router of the destination network.

Q1) On a WiFi, within what will an IP packet be encapsulated?

Answer: Frame

Q2) What do you call the UDP data unit that encapsulates the application data?

Answer: Datagram

Q3) What do you call the data unit that encapsulates the application data sent over TCP?

Answer: Segment

>> Task 7: Telnet

The TELNET (Teletype Network) protocol is a network protocol for remote terminal connection. In simpler words, **telnet**, a TELNET client, allows you to connect to and communicate with a remote system and issue text commands. Although initially it was used for remote administration, we can use **telnet** to connect to any server listening on a TCP port number.

On the target virtual machine, different services are running. We will experiment with three of them:

- **Echo server**: This server echoes everything you send it. By default, it listens on port 7.
- **Daytime server**: This server listens on port 13 by default and replies with the current day and time.
- **Web (HTTP) server**: This server listens on TCP port 80 by default and serves web pages.

Echo and Daytime servers are considered security risks and should not be run; however, we started them explicitly to demonstrate communication with the server using **telnet**.

- ♦ In the terminal, we connect to the target VM at the echo server's TCP port number 7. To close the connection, press the **CTRL +]** keys simultaneously.

Command: **telnet <Machine_IP> 7**

After Connection: Type your Echo_Word/Sentence

Command to quit session: **quit**

- ♦ In the terminal, we use **telnet** to connect to the daytime server listening at port 13. We noticed that the connection closes once the current date and time are returned.

Command: **telnet <Machine_IP> 13**

- ♦ Finally, let's request a web page using **telnet**. After connecting to port 80, you need to issue the command **GET / HTTP/1.1** and identify the host where anything goes, such as **Host: telnet.thm**. Next, you need to press **Enter** twice so your last input line is a blank line. The output below shows the exchange. (The page has been redacted.)

Note: You may have to press **Enter** after sending the information in case you don't get a response.

Command: **telnet <Machine_IP> 80**

After Connection Type: **GET / HTTP/1.1**
Host: telnet.thm

Q1) Use **telnet** to connect to the web server on **Machine_IP**. What is the name and version of the HTTP server?

Answer: lighttpd/1.4.63

Q2) What flag did you get when you viewed the page?

Answer: THM{TELNET_MASTER}